
The Impact of Organizational Commitment on Insiders' Motivation to Protect Organizational Information Assets

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ABSTRACT: Insiders may act to sustain and improve organizational information security, yet our knowledge of what motivates them to do so remains limited. For example, most extant research relies on mere portions of protection motivation theory (PMT) and has focused on isolated behaviors, thus limiting the generalizability of findings to isolated issues, rather than addressing the global set of protective security behaviors. Here, we investigate the motivations surrounding this larger behavioral set by assessing maladaptive rewards, response costs, and fear alongside traditional PMT components. We extend PMT by showing that: (1)

security education, training, and awareness (SETA) efforts help form appraisals; (2) PMT's applicability to *organizational* rather than *personal* contexts depends on insiders' organizational commitment levels; and (3) response costs provide the link between PMT's appraisals. We show in detail how organizational commitment is the mechanism through which organizational security threats become personally relevant to insiders and how SETA efforts influence many PMT-based components.

KEY WORDS AND PHRASES: coping appraisal, information security, MIMIC model, organizational commitment, protection-motivated behaviors, protection motivation theory, security, structural equation modeling, threat appraisal.

Securing organizational information assets is an organization-wide concern that requires increasing financial and personnel resources, because these assets are central to achieving strategic advantage and because both internal and external threats to these assets have increased over time. Organizations use various methods to protect organizational information assets from security threats, including a wide variety of artifacts and efforts: intrusion detection systems; disaster recovery planning; business continuity planning; security education, training, and awareness (SETA) programs; firewalls; and so on. Global information security spending was approximately US\$60 billion in 2012 and is expected to increase to US\$86 billion by 2016 [34]. For example, although the U.S. federal government is decelerating spending on information technology (IT), its spending on IT security is predicted to grow annually by nearly 9 percent between 2011 and 2016 [75]. Despite these major investments, information security continues to challenge even the most seasoned experts as new threats emerge and old threats evolve.

Researchers have found that many managers who oversee security projects overemphasize the use of technology and fail to recognize the importance of human behavior and organizational structures and policies. In actuality, security efforts that account for *organizational insider* (*insider*, for brevity) behavior have the greatest likelihood of success [80]. *Organizational insiders* are full-time employees, part-time employees, temporary workers, and external consultants who have been given authorized access to organizational information [88]. Thus, *behavioral information security research*—the systematic examination of human actions that influence the confidentiality, integrity, and availability of information and information systems [95]—has begun to flourish.

Although technology is vital to organizational information protection, insiders' behaviors ultimately determine the success of security initiatives [25, 69]; technology can extend protection only so far before the control of information must be entrusted to insiders [95]. Moving beyond their initial understanding of insiders as the weakest link in the organizational security chain, researchers are increasingly aware that insiders are equally an often-untapped resource for protecting organizational information assets [1, 28, 80]. Fortunately, evidence indicates that some insiders feel a personal sense of responsibility to protect organizational resources

from security threats [1, 95]. The recognition of this sense of responsibility among researchers represents a considerable shift from the negative view of insiders as the major obstacle to organizational security [16, 17, 21, 22, 36, 43, 100, 109]. We leverage this more positive understanding of insiders as defenders against organizational security threats by examining the role of protection motivation theory (PMT) [84, 85] in understanding insiders' motivation to engage in protective behaviors.

PMT has been used as the foundation for numerous previous efforts in information security [20, 40, 45, 53, 54, 102, 115]. However, we apply PMT in several important ways that have not yet received sufficient attention in the literature. For example, many previous efforts have failed to assess the full PMT nomology or the role of fear in generating protection motivation. Moreover, the vast majority of the published research focuses on the formation of protection motivation (i.e., intentions) without also investigating protective behavior. Our research bridges these gaps by considering the factors of intrinsic and extrinsic maladaptive rewards, response costs, and fear—components central to PMT but rarely explored in the information systems (IS) literature—and their relationships with protection motivation and previously performed protection-motivated behaviors (PMBs) [80] within their organizations.

We also extend PMT in the organizational context in several ways. First, we highlight the importance of SETA initiatives as a major source of security-based information for insiders; such initiatives provide the foundation for the threat and coping appraisal processes specified by PMT. Second, we demonstrate how the relationships proposed by PMT are moderated by insiders' organizational commitment levels and explain why behavioral information security researchers utilizing PMT in organizational contexts must take into account insiders' commitment to their organizations. Finally, we show how the threat and coping appraisal processes are interconnected through insiders' development of response cost perceptions.

Protection Motivation Theory

Overview of Protection Motivation Theory

Although protection motivation theory was applied originally to the field of preventive medicine to explain individuals' protective responses following the communication of health threats via fear appeals [84, 85, 86], PMT is now widely considered a general theory of motivation that can be used to explain individuals' actions regarding any threat. Notably, the objective of fear appeals is not merely to frighten people, but to inspire adaptive, protective behaviors [96]. This communication can be accomplished at both the individual level and the organizational level through direct communication with employees [105]. *Fear* is a potential product of the cost-benefit analysis that occurs once a threat or danger is perceived, and it is a negatively valenced affective state representing a response that arises from a perceived threat and that may include dread, negative arousal, concern or worry, discomfort, or a general negative mood [51, 85, 110].

PMT explains the cognitive processes insiders undergo when faced with threats. These processes motivate insiders to engage in either adaptive or maladaptive responses [86]. *Adaptive responses* are actions that effectively minimize the threat [85]. In such cases, respondents use fear and threat as positive motivators for change—a *danger-control response*. In contrast, *maladaptive responses* are actions that may help to reduce the fear resulting from a threat but fail to actually minimize the threat itself [83]. Such an action is called a *fear-control response*. It is notable, however, that the theoretical assumptions regarding rewards for maladaptive behaviors and response costs for adaptive behaviors have received the least empirical attention [86].

Two appraisal processes are central to PMT: threat appraisal and coping appraisal [86]. In our context, *threat appraisal* is the process by which insiders analyze (1) their perceived threat vulnerability, (2) their perceived threat severity, and (3) potential intrinsic or extrinsic rewards for engaging in maladaptive responses. *Coping appraisal* is the process by which insiders evaluate: (1) the efficacy of the potential adaptive responses to a threat, or *response efficacy*; (2) their ability to successfully carry out the recommended responses, or *self-efficacy*; and (3) the perceived *response costs* associated with their engaging in the adaptive coping strategy [85].

Finally, the outcome of the PMT appraisal processes is a motivational force termed *protection motivation*, which is “an intervening variable that has the typical characteristics of a motive: it arouses, sustains, and directs activity” [85, p. 158]. It drives behavior change and is the lone mediator between the two appraisals and the adaptive responses [85]. Protection motivation should thus be the most significant predictor of future adaptive engagement [67]. Figure 1 provides an overview of PMT.

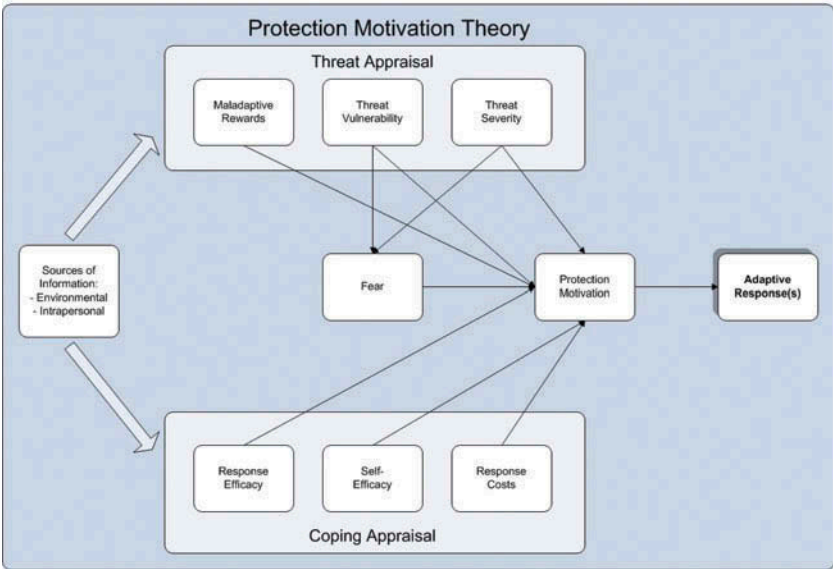


Figure 1. Overview of Protection Motivation Theory

Meta-analyses have demonstrated that PMT's predictions are largely confirmed by empirical findings in several personal contexts [31, 67]. Examples include starting exercise regimens, stopping smoking, conducting breast self-examinations, and receiving regular cervical screenings. PMT research on individual-level information security threats and behaviors has investigated the adoption of home wireless security systems, antispymware and antimalware software, and location-based services.

PMT has also been applied at the organizational level. Welbourne [105] detailed how PMT can be used to guide and accomplish organizational change. Other studies have investigated how to impact organizational change after an initial public offering [106], insiders' reactions to social problems [98], the protection of other insiders [14], the protection of organizations from financial losses [6], and how to thwart information security threats. In the information security field, researchers have used PMT to examine employees' intention to adopt virus protection behaviors [49] and basic protection actions including updating and protecting passwords, updating security and virus software, and backing up system files and documents [9, 115]. In addition, researchers have used PMT at the organizational level to understand insiders' real-world compliance with information security policies [40, 89, 91].

Review of PMT Literature in the Domain of IS Security and of the Open Research Opportunities

We now explain how PMT has been deployed in IS security literature and describe the compelling research opportunities that remain open. Although a sizable number of studies have relied on PMT in this context, its application and consequent results have been inconsistent. Based on our PMT review, we identify multiple ways we could fill these gaps to improve the application of PMT.

First, most PMT studies in IS literature do not consider fear sufficiently, despite its having been introduced as a partial mediator by PMT's pioneers [31, 86]. Several previous research efforts have measured fear with varying results (e.g., [59, 114]), whereas others focus on other facets of PMT such as threat severity, threat susceptibility, and self-efficacy (e.g., [3, 40, 45, 50, 54, 102]). Despite a widespread lack of measurement and assessment, researchers do acknowledge fear as a key part of PMT in their theoretical reviews (i.e., [45, 48]).

Second, few IS PMT studies have tested the complete set of nomological relationships (i.e., the *nomology*) suggested by PMT. The construct of fear is often omitted, as noted above, as are other core PMT constructs. For brevity, we do not catalog every omission, but as previous research has noted [101], the influence of maladaptive rewards and response costs on the development of protection motivation continue to be examined very rarely. A few exceptions exist: one study used intrinsic maladaptive rewards [59], and another focused on the sole extrinsic factor of time savings [102]. Other studies in this area have been conducted [73, 92], but they replace the construct of maladaptive rewards with adaptive rewards in the threat appraisal, which is not suggested by PMT.

Third, although the primary purpose of PMT is to predict protection motivation (i.e., intentions), a natural extension to PMT is to evaluate adaptive, protective behaviors related to the threat and coping appraisals not merely intentions, however noble these may be [31, 86]. Several previous IS PMT efforts have assessed real-world behaviors [37, 44, 48, 73, 90, 92, 113], but most of these studies focus on isolated behaviors, an issue mentioned above.

All these factors combine to form an exciting and fruitful research opportunity in this field. Our study implements a PMT nomology at the organizational level with the goal of providing a comprehensive PMB construct while addressing some other important gaps in the literature.

Explaining Protection-Motivated Behaviors

Information security researchers have examined individuals' security-related activities within various contexts. For example, previous research has investigated individuals' adoption of technologies to protect themselves (e.g., [27, 49]) and their organizations [50] from security threats. Research beyond protective technology adoption has focused on isolated protective intentions or behaviors of adhering to organizational security rules and policies [40, 103]; practicing "safe computing practices" of backing up data [9], changing passwords, refusing to share passwords, scanning e-mails for viruses, updating security software [4, 115]; and exercising general caution with e-mail [69].

Despite the importance of these protective behaviors, they represent only a few of the protective activities that insiders can perform for their organizations [80]. This sparse coverage can impede research on information security [91] because it does not represent protective behaviors as a whole. When researchers examine a single activity or a small subset of behaviors in isolation from a larger structure, the theoretical development of the overall structure is hindered [39].

To understand insiders' protective behaviors more fully, Posey et al. [80] developed a taxonomy of PMBs that encompasses a complete set of beneficial security activities and demonstrated that what is important for organizational security is much more extensive than insiders' passive adherence to the security policies they receive from others. Posey et al. [80] defined PMBs as the volitional behaviors insiders can undertake to protect the following from information security threats: (1) organizationally relevant information within their firms and (2) the computer-based systems in which that information is stored, collected, disseminated, or manipulated. Based on the field of systematics and its science of diversity, their taxonomy classifies individual PMBs on the basis of their similarities; however, subsequent systematic research on the science of universals (i.e., the theoretically derived antecedents influencing a phenomenon of interest predefined by the science of diversity efforts) on the overall PMB construct is notably lacking. This limitation prevents researchers from fully examining the primary factors that motivate insiders to adapt their own roles in order to protect their organizations from information security threats.

Two points regarding PMBs are worth noting at this point. First, we assert that PMBs are insiders' volitional activities [80]. Insiders have substantial control over the information they are exposed to in their jobs [95], yet whether to protect this information actively is typically their choice. Insiders may also perform these actions regardless of any potential sanctions for not protecting information.

Second, insiders may expend more effort to engage in certain PMBs than in others [80]. Certain behaviors may require insiders to use their best judgment about what constitutes an information security threat. For example, properly logging in and out of the computer systems in the workplace after completing job tasks is straightforward. However, reporting a coworker's negligent IT actions to management demands effort, carries risk, and offers an uncertain outcome. Insiders are often reluctant to blow the whistle on the misdeeds of peers due to fear of retaliation by a coworker or fear of corroding general morale [57]. Intra- and interpersonal variations exist among PMB motivations. Some insiders are more motivated overall to engage in PMBs, but there is considerable variation among the types of PMBs in which they may be motivated to engage.

Theoretical Model Based on Protection Motivation and Engagement

In this section, we propose a theoretical model that first builds on PMT and then systematically extends it to leverage the opportunities, identified in the literature review, to explain why insiders become motivated to engage in PMBs. Further, we also show how PMT can be used to explain previously performed PMBs, as clearly organizations do not just want to create protection intentions but in actuality, protective behaviors. Figure 2 displays our conceptual model. We use the remainder of this section to develop our hypotheses, including justifications for our proposed extensions for organizational commitment and SETA programs.

SETA Programs as an Antecedent to Protection Motivation

SETA programs typically help organizations with risk mitigation strategies, enhanced security stature, and the ability to protect valuable corporate assets. The goal of a SETA program is to reduce the organization's security risk by reducing accidental security breaches and increasing organizational resistance to other forms of attack. SETA programs consist of three security elements: education, training, and awareness [107]. Each of these elements should be ongoing for the organization, because information security is a highly dynamic phenomenon. SETA programs should accomplish several tasks, including [23]: (1) communicating knowledge about organizational information threats and risks, (2) explaining existing technical and procedural countermeasures available to employees, (3) detailing possible sanctions imposed by the organization for security policy violations, and (4) increasing employees' awareness of their roles and responsibilities in protecting the organization's information assets.

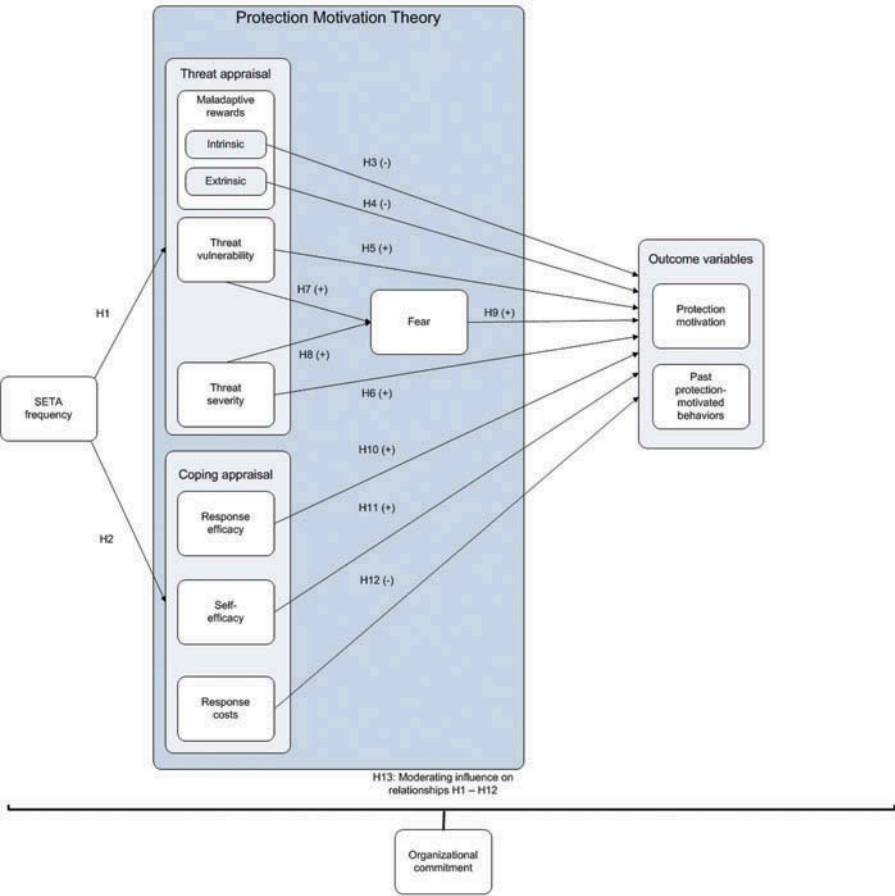


Figure 2. Conceptual Model of Protection Motivation and Engagement

Researchers have found that SETA programs are useful for guiding insiders’ intentions and behaviors regarding organizational information assets [23, 24]. Appropriate SETA programs clearly communicate information security threats and coping behaviors for the majority of possible threats. For this reason, we believe SETA programs already act as the key distribution channel of PMB-related fear appeals within organizations, and are thus useful antecedents to PMT’s threat and coping appraisal processes. The key to getting insiders to engage actively in PMBs is to balance positive and negative information. Schein [87] explained that for positive change to occur, insiders need to have a manageable path forward that helps them appreciate new or improved behavior and that provides appropriate direction and support. For this reason, fear appeals must be balanced with coping information to gain a transformational result, and SETA programs accomplish this nuanced task. Under PMT, an insider can

receive information regarding an organizational security threat from a variety of environmental and intrapersonal sources. In addition to providing effective explanations of individuals' responses to individual threats, PMT can also be used to explain how to thwart multiple organizational security threats simultaneously through adherence to information security policies [40, 91, 102]. *External sources* of threat information include discussing a security problem encountered in a system with coworkers, observing how other insiders deal with sensitive organizational information when traveling, learning of security breaches in the news, and so on. *Internal sources* of information used in the PMT process include one's personality, previous experience, and lessons learned from previous coping activities.

Formal SETA programs serve as significant external sources of security threat information for insiders [24, 107]. Organizational spending on SETA programs is expected to remain steady or increase over time due to their importance [29]; however, considerable variation exists in their content, delivery, and frequency. Given that a lack of regular SETA programs is cited as a primary explanation for the ineffectiveness of plans for threat responses within organizations—that is, insiders “who don't know how to do things rarely do them well” [81, p. 17]—SETA programs provide the foundation upon which insiders are able to gauge many of the factors regarding information security threats accounted for in the appraisals specified by PMT. When performed regularly, SETA programs remind insiders of security threats faced by their organization, their role in the battle against these threats, and why the organization is being targeted by the threats. We thus hypothesize:

Hypothesis 1: Insiders who receive more frequent instruction via SETA programs in their organizations will have an improved understanding of PMT's threat appraisal process.

Hypothesis 2: Insiders who receive more frequent instruction via SETA programs in their organizations will have an improved understanding of PMT's coping appraisal process.

Predictions from PMT's Basic Assumptions and Nomology

According to PMT, after an insider acquires security threat information, the insider is able to evaluate the security threat via the threat appraisal process. First, the insider assesses any potential rewards or personal gains for not engaging in the PMBs that address the security threats. These negatively earned gains are termed *maladaptive rewards* [86]. Maladaptive rewards can be *intrinsic* or *extrinsic*. The former might come, for example, from the potential satisfaction of allowing the organization to be harmed via security threats in the event that the insider is demoralized and seeks indirect retaliation, whereas the latter might take the form of potential financial rewards from outsiders who are seeking corporate trade secrets or even trying to

provide limited information to help a friend outside of work. If PMT holds in our context, such perceived rewards will have a negative influence on insiders' motivation to engage in future PMBs as well as the degree by which PMBs have been previously performed in the organization.

Hypothesis 3: Increased intrinsic maladaptive rewards for not performing PMBs decreases insiders' (a) protection motivation and (b) engagement in PMBs.

Hypothesis 4: Increased extrinsic maladaptive rewards for not performing PMBs decreases insiders' (a) protection motivation and (b) engagement in PMBs.

Despite potentially alluring high maladaptive rewards, insiders might still choose to engage in PMBs because they understand the potential damage their organizations could suffer should the threats be realized. Employees spend nearly half of their waking lives engaged in work activities and often become attached to their organizations, including the organizations' goals and stakeholders (e.g., [77]). This positive connection between employees and organizations often leads to an increased frequency of beneficial activities performed by employees on behalf of their organizations [18, 99]. This connection is why many insiders feel responsible for protecting organizational information resources from security threats [1, 95].

Thus, *threat vulnerability* should be a major component in the threat appraisal process and overall formation of insiders' protection motivation. In our context, *threat vulnerability* is the extent to which insiders feel that their organizations are susceptible to a particular threat or that the threat is probable [58]. *Threat severity*—the extent to which organizational threats are perceived by insiders to be detrimental and to cause harm [58]—should also influence insiders' protection motivation. When insiders perceive that their organizations are vulnerable to security threats, insiders should become more motivated to protect their organizations, assuming that most insiders feel loyalty to and have a positive connection with their organizations. Security threats viewed as more harmful should also heighten these feelings of personal responsibility to engage in PMBs.

Hypothesis 5: Increased perceived threat vulnerability regarding their organizations' information security threats increases insiders' (a) protection motivation and (b) engagement in PMBs.

Hypothesis 6: Increased perceived threat severity regarding their organizations' information security threats increases insiders' (a) protection motivation and (b) engagement in PMBs.

When individuals assess threat vulnerability along with threat severity, they often feel afraid [84]. The combination of inevitable events that inflict discomfort often causes individuals to become nervous, scared, and upset [86]. Although the revised PMT model [85] does not include a direct link between fear and protection motivation, other researchers have argued that fear is a necessary component of the cognitive mediating processes suggested by PMT and should receive greater

consideration [30, 97]. Consequently, in later PMT revisions, fear was situated in a partial mediation role between threat and protection motivation [31, 86]. If PMT holds in our context, then insiders' perceptions of the security threats encountered will influence the degree of fear they experience. This fear should also influence the degree to which insiders are motivated to protect their organizations from those security threats in the future as well as explain employees' engagement in protective responses in the past.

Hypothesis 7: Increased perceived threat vulnerability regarding their organizations' information security threats increases insiders' fear.

Hypothesis 8: Increased perceived threat severity regarding their organizations' information security threats increases insiders' fear.

Hypothesis 9: Increased fear related to their organizations' information security threats increases insiders' (a) protection motivation and (b) engagement in PMBs.

The second PMT process is the coping appraisal process [85], which meta-analysis has shown to be more influential in forming protection motivations than the threat appraisal process [67]. This process involves the consideration of response efficacy, self-efficacy, and response costs. *Response efficacy* is the perception that the recommended coping strategies can successfully attenuate the threat [83], and has been shown to play a vital role in forming insiders' protection motivations [40]. Some researchers have maintained that response efficacy is the most important predictor of protection motivation [8, 112].

Hypothesis 10: Increased response efficacy regarding PMBs increases insiders' (a) protection motivation and (b) engagement in PMBs.

The revised version of PMT [85, 86] includes self-efficacy [5] within the coping appraisal process. *Self-efficacy* is the belief that an individual is personally capable of implementing the proposed coping strategy appropriately, and has been shown to strongly predict protection motivation in a wide range of contexts [40, 67].

Hypothesis 11: Increased self-efficacy regarding PMBs increases insiders' (a) protection motivation and (b) engagement in PMBs.

The final component of the coping appraisal process, *response costs*, constitutes insiders' perceived drawbacks for engaging in protective actions [86]. These costs include any expenses, inconveniences, difficulties, and potential side effects that insiders believe they will incur from performing protective actions [33]. Just as maladaptive rewards influence the threat appraisal process, response costs decrease the likelihood that insiders will perform adaptive responses [74].

Hypothesis 12: Increased response costs regarding PMBs decrease insiders' (a) protection motivation and (b) engagement in PMBs.

Extending PMT in Organizational Settings with Organizational Commitment

Much research has focused on why and how employees become connected to their organizations and how these connections influence motivational levels and consequent important workplace behaviors. Because insiders may become committed to organizations for various reasons, three major types of organizational commitment have been identified: affective, continuance, and normative [63]. Employees with high *affective organizational commitment* are those who want to continue their organizational membership because the organization's values, goals, and initiatives align with the employees' views [63]. Employees with high *continuance organizational commitment* stay with organizations simply because the costs of leaving it are too great and the alternatives provide no greater benefit [63]. Finally, employees with high *normative organizational commitment* feel obliged to continue as organizational members because they are expected to or believe they have already invested too much time to leave [63].¹

Affective organizational commitment is appropriate in the context of PMT because insiders with high levels of this commitment should embrace the initiatives and views of the organization. Affective organizational commitment has received the most attention in the academic literature because of its importance in driving beneficial behaviors and because employees with high affective commitment are those that organizations desire to retain [63]. For example, employees with high affective commitment have been shown to perform at higher levels in their positions than their counterparts with lower affective commitment [63], and the former are also more likely to respond positively to negative information [64]. Previous research has shown that employees with high affective commitment to their organizations are also more likely to engage in citizenship behaviors, that is, positive actions that involve "going the extra mile" rather than merely fulfilling explicit job requirements [71], more likely to follow organizational policy [46], and more likely to report the negligent activities of their coworkers (e.g., whistle-blowing) [93].

These and other important relationships exist within the organizational context because those with high affective commitment to their organizations are emotionally attached to, identify with, and desire to be involved with the organizations and are thus willing to do their best to support their organizations' goals [61, 68]. Moreover, insiders with high affective commitment view their organizations' values as congruent with their own; thus, when one party succeeds, so does the other. It is this commitment that is a major "energizing force" for motivated organizational behavior [65, p. 993]. Thus, affective commitment should make security threats to insiders' organizations more relevant to them.

Because the core purpose of PMT is to foster high levels of protection motivation so that threats are averted or their effects mitigated, we assert that insiders with high affective organizational commitment will respond differently to information security threats affecting the organization than their less committed colleagues. We expect that many of these differences will be exhibited from the initial stages of appraisal formation during information acquisition to the development of protection

motivation itself. The extant literature provides much support for this assertion, although primarily through exploratory research. The first study to suggest the link between organizational commitment and positive behaviors (e.g., ethical behaviors) in IS employees was [72], but it offered little theoretical explanation for this link. Another study [94] went further and showed a positive relationship between organizational commitment and protective security behaviors in organizations. Because the research was atheoretical and exploratory, they could only speculate that this was the case because such employees are more engaged in productive work and have less time to engage in risky behaviors (e.g., surfing the web) or because they may take admonitions from security personnel more seriously than less committed employees. Herath and Rao [41] built insightfully on this work in a larger PMT/deterrence theory framework, and they further explained and provided empirical support for two more specific premises regarding organizational commitment: that it improves employees' perceptions of the effectiveness of their actions (or their response efficacy) and that it increases the likelihood of their following security policies. Another study [52] took a more indirect approach and explained that the combination of identifying with an organization and being exposed to its norms furthers the development of positive prosocial norms that lead to policy compliance intentions.

It follows from these studies [41, 52] that insiders who are closely committed to their organizations should be more likely to be attentive and thoughtful when engaging in SETA programs, because they desire to acquire information about how best to protect the organization to which they are so deeply committed. Consequently, these positive behaviors are fostered as key foundations of their work norms. They would thus see conforming to SETA programs as part of their jobs and a way to engage in productive work, and they would also more likely experience an adaptive threat and coping appraisal process: that is, they would better recognize and pay attention to threats, and they would likewise respond in a more efficacious and adaptive manner. Less committed employees do not see their values as congruent with those of their organizations and are much less likely to exert effort, especially extra effort, in protecting them, because the threats are not perceived as personally relevant. They would likewise receive less efficacy from SETA programs, because they would be less likely to make such programs internally relevant.

Consequently, we view affective organizational commitment as the "missing link" in applying PMT to information security studies within organizational settings. Although we have just reviewed the previous literature incorporating organizational commitment as an antecedent to insiders' security-related intentions and behaviors, we wish to expand on it significantly by asserting the role of organizational commitment as a moderator of the entire PMT process. Crucially, if research efforts include employees of both low and high organizational commitment, the relationships suggested by PMT should be skewed, because the two employee groups do not view the organization in similar fashions; not accounting for differences in organizational commitment could undermine, perhaps critically, the interpretation of PMT organizational security studies. We thus propose and test the moderating influence of affective commitment on our overall conceptual model.

Hypothesis 13: Insiders with high levels of affective organizational commitment will be more strongly affected by the components of PMT than insiders with low commitment levels.

Methodology

Data Collection Approach

We collected data for this study using a survey panel of insiders requisitioned by a panel provider. The provider compensated the panelists for their participation. Panels have been used to elicit responses to survey instruments in various settings, including the organizational security context [57, 80]. Such panels offer several advantages for organizational security research. First, panels guarantee anonymity to the respondent, a necessary element in eliciting honest responses to questions about behaviors potentially influenced by social desirability beliefs [7]. Second, respondents from a wide range of industries and positions can be accessed for topics requiring the participation of a broad spectrum of insiders who would be nearly impossible to include otherwise. Finally, because of the sensitive nature of information security, organizations are less likely to allow outside researchers to gain access to employees [47].

Sample of Professional Organizational Insiders

Out of 570 panelists who clicked a link to consider participating in our survey, 380 completed the survey, giving us a completion rate of approximately 67 percent. The sample consisted of 380 insiders from various industries and positions within the United States. The sample was 53.4 percent female, 10.5 percent IS or IT professionals, and 34.6 percent managers; 96.1 percent of the respondents held full-time positions. The average age was 43.75 years, and the average portion of a typical working day spent using their organizations' computer systems was reported to be 65.4 percent.²

Rival Explanations and Demographic Controls

As with all research relying upon a single theory, testing and controlling for rival explanations is important for understanding as fully as possible the potentially significant influences on an organizational phenomenon not specified by the theory. The goal is to isolate the influence of the theoretical model to make sure it is actually predicting the dependent variable, in our case PMBs. These rival explanations should be considered controls that isolate the influence of our theoretical PMT model. Thus, we examined the IS and management literatures extensively for alternative motivating factors that might have a significant influence on insiders' protection motivation. Our search highlighted three potential rival explanations for positive employee motivations and behaviors: job satisfaction, extrinsic financial incentives, and management support.

First, research has established a positive relationship between satisfaction and employee in-role and extra-role organizational behaviors (e.g., [108]). As previously noted, PMBs comprise both in-role and extra-role behaviors, and satisfied employees will thus have a greater propensity to exhibit these behaviors than unsatisfied employees. Second, financial incentives have always been used to influence employees' activities and behaviors, and the performance of PMBs is no exception to this rule. Finally, management support is central to accomplishing major tasks and changes within organizations. Hu et al. [42] detailed the importance of top management support with respect to information security policy compliance and organizational culture change. We also included the following demographic controls: gender, age, managerial status, degree of computer usage at work, organizational tenure, and IT-IS professional status.

Construct Measurement

We used previously validated scales and adapted them to this study where available. Unless otherwise indicated, we collected responses on a seven-point Likert-type scale (1 = strongly disagree; 2 = moderately disagree; 3 = slightly disagree; 4 = neither disagree nor agree; 5 = slightly agree; 6 = moderately agree; 7 = strongly agree). Table 1 provides a high-level summary of the measures. SETA frequency

Table 1. Summary of Construct Measurement

Measure	Number of items	Cronbach's alpha	Citation
Intrinsic maladaptive rewards	2	0.73	Based on McClendon et al. [60]
Extrinsic maladaptive rewards	3	0.65	Based on O'Driscoll and Randall [70]
Threat vulnerability	4	0.90	Witte et al. [111]; Workman et al. [115]
Threat severity	4	0.90	Witte et al. [111]; Workman et al. [115]
Fear	6*	0.94	Block and Keller [8]
Response efficacy	3	0.85	Workman et al. [115]
Self-efficacy	3	0.80	Workman et al. [115]
Response costs	4	0.85	Workman et al. [115]
Protection motivation	3	0.64	Rogers [85]; Rogers and Prentice-Dunn [86]; Tanner et al. [96]
Past protection-motivated behaviors (PMBs)	5	0.87	Authors, blinded
Affective organizational commitment	6	0.85	Meyer and Allen [63]
Job satisfaction	3	0.90	Cammann et al. [12]
Financial incentives	3	0.78	O'Driscoll and Randall [70]
Managerial support	4	0.83	Campion et al. [13]

* = measured on a five-point scale.

was modeled as a categorical variable. Because insiders experiencing no rather than some formal SETA programs within their organizations are likely to have significantly different experiences, we first grouped these individuals by themselves (0 = no formal SETA programs, $n = 71$). The remaining individuals were grouped according to the median of the SETA frequency data (1 = once or twice per year, $n = 112$; 2 = more than twice per year, $n = 197$). Gender was also modeled categorically (0 = female; 1 = male), as were managerial status (0 = not managerial; 1 = managerial) and IT-IS professional status (0 = not IT-IS; 1 = IT-IS). See Appendix 1 [online supplemental data] for measurement details. We used the Block and Keller [8] measure for fear because it provided the most global self-reported measure available. Self-reported fear, as measured by mood adjectives, adequately captures fear, because it includes the correspondence between physiological arousal and self-ratings of mood adjectives. In fact, they argued that self-rated fear is more global in nature and more adequately reflects an overall emotional state, whereas physiological arousal fluctuates substantially during the presentation of a fear appeal [85, 110].

Importantly, the measures were highly contextualized to security-using professional experts. To validate that the measurement items had relevance in organizational contexts, the first author interviewed 11 information security professionals and 22 traditional insiders to gather information about the components suggested by PMT (Authors blinded, 2014). Because the overall set of PMBs identified by Posey et al. [80] is intended to be representative of protective behaviors in most organizations across different employment situations, the professionals and insiders were selected from a wide variety of industries, including the financial, insurance, legal, military, telecommunications, aviation, and medical fields, and different levels of experience. The data relative to each PMT component were then compared to the relevant measurement items.

Finally, Appendix 2 [online supplemental files] details information about the establishment of the reflective PMB scale and how it was assessed relative to the unique behaviors discovered by Posey et al. [80]. Briefly, this assessment was accomplished via a multiple indicators and multiple causes (MIMIC) model, which is a construct modeled to have both formative and reflective components [15, 26, 79]. This examination aided us in determining the degree to which the overall reflective PMB measure captures the concept domain covered by the individual insider activities. The formative behaviors explained over 70 percent of the variance of the overall PMB measure, thereby indicating considerable coverage of the domain.

Analysis

We tested our theoretical model (see Figure 2) using the covariance-based structural equation modeling (SEM) program Mplus version 7. We chose Mplus both because it is a covariance-based technique that allows the entire PMT theory to be assessed

for model fit and because it has the ability to handle data that depart from normality. Accordingly, we used the maximum likelihood estimation with robust standard errors (the MLR estimator option in Mplus) in our assessments. Notably, previous IS PMT security research using partial least squares regression (PLS) analysis could not test the model fit of PMT, because model fit statistics cannot be calculated with PLS [56]. We followed the two-step process of examining separate measurement and structural models, and we discuss that process in the following sections.

Measurement Model and Construct Validity

The first step in assessing the unmoderated hypothesized model (without H13) was to perform a confirmatory factor analysis (CFA). The initial CFA model with all multi-item latent constructs, including rival explanations, indicated a few potential issues. All items loaded on their respective constructs with a highly significant *t*-value ($p < 0.001$), but several items exhibited standardized regression weights of less than 0.60, so we removed those items. We also analyzed the standardized residual covariance matrix to assess other potentially problematic items within the model. Items exhibiting significant values in this matrix (2.58 or greater) were considered for removal. We removed two items (Fear3, Fear6) that yielded undesirable standardized residuals.

We used three criteria to assess the convergent validity of all the reflectively modeled constructs: (1) factor loadings, (2) average variance extracted (AVE), and (3) internal consistency estimates. As stated, all remaining factor loadings were highly significant and above the 0.60 cutoff value. We calculated AVE values for each construct (see Table A1.3 in [online] Appendix 1). All the constructs had AVEs greater than the 0.50 heuristic. In addition, all the constructs exhibited internal consistency, with Cronbach's alpha coefficients greater than 0.70, except for slightly lower alphas for protection motivation and extrinsic maladaptive rewards, and thereby met the demands set forth by previous research. Due to the manner in which alpha scores are calculated, lower alpha scores may be experienced with instruments having few items.

We assessed discriminant validity per the guidelines developed by Fornell and Larcker [32], which require that constructs in a measurement model maintain discriminant validity if the square root of the AVEs of both constructs under consideration is higher than the correlation between those two constructs. As shown in Table A1.3, two paired correlations did not meet this criterion. Threat vulnerability and threat severity exhibited an association of 0.831, and the correlation between response efficacy and self-efficacy was 0.845. Although such high correlations are not uncommon in research and do not necessarily preclude analysis via SEM techniques [35], the square root of the AVEs for threat vulnerability and self-efficacy were not high enough to justify discrimination, and we had to discard these constructs from the model. In addition, we chose to keep response efficacy rather than self-efficacy and threat severity rather than threat vulnerability in the

model because previous meta-analytic findings regarding PMT research [83, 112] have demonstrated that they exhibit stronger relationships with protection motivation than do the alternatives. Other notable correlations include those between intrinsic and extrinsic rewards ($r = 0.559$), response efficacy and past PMBs ($r = 0.529$), protection motivation and past PMBs ($r = 0.602$), and management support and protection motivation ($r = 0.521$), although none of these warrant concern with respect to discriminant validity. Thus, all remaining constructs met the third criterion for discriminant validity, and the overall construct validity was established. The refined CFA model fit the data well with a $\chi^2 = 637.192$; $df = 505$; scaling correction factor = 1.0822; CFI = 0.978; and RMSEA = 0.026.

As a final examination of the measurement model, we assessed whether common methods bias significantly attributed to the variance exhibited among the survey items. For this assessment, we utilized the unmeasured latent methods or “marker” construct (ULMC) approach, wherein a latent construct is added to the CFA model that is an aggregate of all the manifest items used in the study [82]. Any significant variance explained by this marker construct can be attributed to methods bias. Fortunately, when comparing the standardized loadings of the items on their respective constructs between CFAs with and without this marker construct, the average difference across all items’ standardized loadings was less than 0.020, with a maximum difference of 0.067. Furthermore, none of the items loaded significantly on the marker construct; thus, we can conclude that common methods bias was not likely to be present at any significant level in our data.

Structural Model

After we validated the constructs and obtained an acceptable fit to the data set’s covariance matrix via CFA, we tested the hypotheses by converting the CFA model into a structural model. Of the rival explanations and demographic controls, only managerial status (0 = no; 1 = yes) and managerial support were significant in explaining variance in protection motivation. This outcome was true for the overall group and both subgroups, as detailed below. Thus, the other rivals and controls were removed from further analyses. The final CFA for the entire data set with financial rewards and job satisfaction removed exhibited the following statistics: $\chi^2 = 479.905$; $df = 369$; scaling correction factor = 1.0880; CFI = 0.977; RMSEA = 0.028.

The unmoderated structural model including the categorical variables of SETA frequency and managerial status exhibited the following characteristics: $\chi^2 = 867.973$; $df = 442$; scaling correction factor = 1.0861; CFI = 0.913; and RMSEA = 0.051, with a 90 percent confidence interval of 0.046–0.056. This borderline fit of the conceptual model to the data was puzzling given the relatively high fit statistics from the CFA. Upon a review of the modification indices provided by Mplus, however, we noted a strong interplay among the constructs not originally stated by Rogers [85] and potentially hidden from other assessments of PMT using component-based SEM techniques (i.e., PLS). These findings indicated the need to estimate the relationships

between intrinsic maladaptive rewards, extrinsic maladaptive rewards, and response efficacy with response costs. We discuss the reasons why these relationships are meaningful in the discussion section, but here we briefly note that individuals' formation of response costs is a cost–benefit analysis, wherein benefits are linked to potential rewards and costs to whether the proposed actions are thought to equate to salient change. Once these discovered relationships were estimated in the revised structural model, the model exhibited the following, more appropriate characteristics: $\chi^2 = 787.688$; $df = 439$; scaling correction factor = 1.0834; CFI = 0.929; and RMSEA = 0.046, with a 90 percent confidence interval of 0.041–0.052. The revised structural model indicates an acceptable fit to the data set, given the model's complexity [38].

Assessing Organizational Commitment's Moderating Effect

To assess H13, we followed [9] and divided the respondents based on their median scores on affective organizational commitment so that those below the median (a score of 5) were placed in the “low” group ($n = 182$) and those at or above the median were placed in the “high” group ($n = 198$). The basic statistics exhibited by affective organizational commitment were as follows: mean = 4.76; $\sigma = 1.50$; and $\alpha = 0.86$. The measurement models with multi-item latent constructs exhibited acceptable fit statistics for each group: low group— $\chi^2 = 490.316$; $df = 369$; scaling correction factor = 1.0100; CFI = 0.956; RMSEA = 0.043; and high group— $\chi^2 = 453.750$; $df = 369$; scaling correction factor = 1.0392; CFI = 0.962; RMSEA = 0.034. Before comparisons could be made between the two groups' structural models, however, it was imperative to test whether the two groups' measurement models were invariant, that is, whether respondents in the two groups did not differ in how they understood the survey items (aka, *metric invariance*) [11].

We first assessed full metric invariance, the most stringent form, in which all loadings on reflectively modeled constructs are equal between the groups of interest, but were unable to establish evidence of invariance in this manner. Because full metric invariance is rarely established in real-world data collections, we then chose to assess partial metric invariance before testing H13. Partial metric invariance stipulates that as long as two of the loadings per construct are invariant or are not significantly different from one another, structural comparisons can be made [11]. Accordingly, we randomly selected which two loadings—except the loading constrained to one, which is used to set the scale for the construct—would be held invariant between the low and high groups for each construct in the model. When fewer than four items existed for the construct, all loadings were held equal between the groups on that construct: intrinsic and extrinsic maladaptive rewards, response efficacy, protection motivation, and management support.

Our assessment of partial metric invariance demonstrated that the two groups understood the multi-item survey constructs similarly.³ The following statistics were obtained: constrained model— $\chi^2 = 964.995$; $df = 753$; scaling correction factor = 1.0314; and baseline model— $\chi^2 = 943.545$; $df = 738$; scaling correction factor =

1.0246. After we corrected for the scaling differences, we performed the χ^2 difference test. A corrected $\Delta\chi^2$ of 20.893 with $\Delta df = 15$ yielded a probability of 0.140, thereby providing evidence of partial metric invariance and allowing comparisons between the two structural models.

We also calculated effect sizes (f^2 scores) as well as pseudo- f tests for the overall group and both moderator groups for both dependent variables in our model. Both calculations provided evidence of the importance of the PMT-based components in explaining the variance in protection motivation and past PMBs above that explained by the controls. For protection motivation: the overall group, $f^2 = 0.074$ and pseudo- $f = 27.37$ ($df = 371$); the low organizational commitment group, $f^2 = 0.058$ and pseudo- $f = 10.05$ ($df = 173$); the high organizational commitment group, $f^2 = 0.534$ and pseudo- $f = 100.35$ ($df = 189$). For past PMBs: the overall group, $f^2 = 0.229$ and pseudo- $f = 84.62$ ($df = 371$); the low organizational commitment group, $f^2 = 0.117$ and pseudo- $f = 20.09$ ($df = 173$); the high organizational commitment group, $f^2 = 0.642$ and pseudo- $f = 120.76$ ($df = 189$). All pseudo- f scores exhibited a statistical significance of $p < 0.001$.

Discussion of Results

PMT Results

Although we did not find support for the relationship between SETA frequency and intrinsic maladaptive rewards (H1a is not supported), we did find support for SETA frequency's relationship with both extrinsic maladaptive rewards and threat severity within the threat appraisal portion suggested by PMT (H1b and H1d are supported). For the coping appraisal components, SETA frequency exhibited a significant relationship with response efficacy (H2a is supported) but not response costs (H2c is not supported). Intrinsic maladaptive rewards exhibited a significant negative relationship with insiders' protection motivation and past PMBs (H3a and H3b are supported), whereas extrinsic maladaptive rewards displayed an insignificant negative relationship with both dependent variables (H4a and H4b are not supported). Threat severity failed to demonstrate a significant relationship with protection motivation (H6a is not supported), but it did explain significant variance in past PMBs and fear (H6b and H8 is supported). Fear was not significantly related to protection motivation or past PMBs (H9a and H9b are not supported).

In the coping appraisal process, response efficacy exhibited strong positive relationships with protection motivation and past PMBs (H10a and H10b are supported). Although we were not able to test self-efficacy's relationship with the two dependent variables (H10a and H10b), we were able to examine response costs' relationship with protection motivation and past PMBs; the structural model demonstrates a significant negative relationship in this regard with protection motivation (H12a is supported) but not past PMBs (H12b is not supported). [Table 2](#) summarizes the complete findings of this assessment.

Table 2. Results from the Revised Conceptual Model

Tested paths	β	t -statistic
<i>Relationships derived from base PMT</i>		
H3a: Intrinsic maladaptive rewards $\rightarrow$ (–) Protection motivation	(–0.191)	–1.994*
H3b: Intrinsic maladaptive rewards $\rightarrow$ (–) Past PMBs	(–0.142)	–2.184*
H4a: Extrinsic maladaptive rewards $\rightarrow$ (–) Protection motivation	0.041	0.776 (n/s)
H4b: Extrinsic maladaptive rewards $\rightarrow$ (–) Past PMBs	0.045	1.041 (n/s)
H6a: Threat severity $\rightarrow$ Protection motivation	0.062	0.908 (n/s)
H6b: Threat severity $\rightarrow$ Past PMBs	0.101	2.051*
H8: Threat severity $\rightarrow$ Fear	0.290	5.057***
H9a: Fear $\rightarrow$ Protection motivation	0.086	1.392 (n/s)
H9b: Fear $\rightarrow$ Past PMBs	0.035	0.647 (n/s)
H10a: Response efficacy $\rightarrow$ Protection motivation	0.236	3.207***
H10b: Response efficacy $\rightarrow$ Past PMBs	0.441	7.211***
H12a: Response costs $\rightarrow$ (–) Protection motivation	(–0.190)	–2.311*
H12b: Response costs $\rightarrow$ (–) Past PMBs	(–0.104)	–1.481 (n/s)
<i>Relationships derived from rival explanations or controls</i>		
Management support $\rightarrow$ Protection motivation	0.373	5.684***
Management support $\rightarrow$ Past PMBs	0.135	2.279*
Managerial position $\rightarrow$ Protection motivation	0.116	2.186*
Managerial position $\rightarrow$ Past PMBs	0.153	3.642***
<i>Extensions to base PMT model</i>		
H1a: SETA frequency $\rightarrow$ Intrinsic maladaptive rewards	0.026	0.494 (n/s)
H1b: SETA frequency $\rightarrow$ Extrinsic maladaptive rewards	0.110	2.082*
H1d: SETA frequency $\rightarrow$ Threat severity	0.180	3.391***
H2a: SETA frequency $\rightarrow$ Response efficacy	0.316	5.891***
H2c: SETA frequency $\rightarrow$ Response costs	0.078	1.457 (n/s)
†Intrinsic maladaptive rewards $\rightarrow$ Response costs	0.301	5.208***
†Extrinsic maladaptive rewards $\rightarrow$ Response costs	0.142	2.443*
†Response efficacy $\rightarrow$ Response costs	(–0.392)	–6.266***
Variance explained	R^2	Significance
Intrinsic maladaptive rewards	0.001	0.247 (n/s)
Extrinsic maladaptive rewards	0.012	1.041 (n/s)
Threat severity	0.032	1.695 (n/s)
Response efficacy	0.100	2.945**
Response costs	0.249	4.017***
Fear	0.084	2.529*
Protection motivation	0.351	4.699***
Past PMBs	0.331	5.694***

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, n/s = nonsignificant; † = relationships suggested by modification indices; relationships concerning threat vulnerability and self-efficacy could not be assessed.

PMT Extensions Results

Mplus identified other potential extensions to our conceptual model via modification indices. These additions included estimation of the relationships between (1) intrinsic and extrinsic maladaptive rewards with response costs and (2) response efficacy with response costs. All were significant at the 0.05 level of significance or lower.

PMT Moderation Results

As shown in Table 3, we found mixed support for H13. In fact, we found several important relationships that are moderated by the level of affective organizational commitment. Of these, the most prominent was the relationship between response efficacy and protection motivation, followed by the relationships between intrinsic maladaptive rewards with response costs and threat severity with protection motivation and past PMBs. The remaining important relationships were H1a and H1b, which represents SETA frequency's association with intrinsic and extrinsic maladaptive rewards. Table 3 displays information regarding our assessment of H13 for the original conceptual model, as well as its extensions.

Using Cohen's [19] suggestions for determining the strength of effect sizes of changes in R^2 values (0.02 = small; 0.15 = medium; 0.35 = large), we found that for the overall group, the addition of PMT to the controls-only model resulted in a small effect of 0.074 ($R^2_{\text{partial}} = 0.303$; $R^2_{\text{full}} = 0.351$) for protection motivation and a medium effect of 0.229 ($R^2_{\text{partial}} = 0.178$; $R^2_{\text{full}} = 0.331$) for past PMBs, whereas for the low-commitment group this addition does considerably less to explain variance in protection motivation ($R^2_{\text{partial}} = 0.221$; $R^2_{\text{full}} = 0.264$; effect = 0.058) and past PMBs ($R^2_{\text{partial}} = 0.149$; $R^2_{\text{full}} = 0.238$; effect = 0.117) beyond the controls. However, when considering the findings for the high-commitment group, the addition of PMT to the controls-only model resulted in very large effect sizes of 0.534 ($R^2_{\text{partial}} = 0.319$; $R^2_{\text{full}} = 0.556$) and 0.642 ($R^2_{\text{partial}} = 0.100$; $R^2_{\text{full}} = 0.452$) for protection motivation and past PMBs, respectively. We can thus conclude that PMT and its components are much more useful for explaining the cognitive, motivational, and previous behavioral patterns of insiders for those with high organizational commitment than they are for explaining such patterns for those with low commitment.

Contributions to Research, Theory, and Practice

This study offers several findings that make significant contributions to both practitioners and researchers who are developing theories and gathering empirical data related to information security. Specifically, we have made the following contributions to IS theory-building in PMT: (1) we built on the nomology of PMT; (2) we tested fear as a potential partial mediator in the model; (3) we tested the baseline of PMT and our extensions using covariance-based SEM in order to report both overall

Table 3. Results from Organizational Commitment Moderation of the PMT Model

Tested paths	Overall		Low org. commit.		High org. commit.	
	B		B		B	
<i>Testing the Normology of the Baseline PMT Model</i>						
H3a: Intrinsic maladaptive rewards → (–) Protection motivation	(–0.191)*		(–0.091)		(–0.362)	
H3b: Intrinsic maladaptive rewards → (–) Past PMBs	(–0.142)*		(–0.162)		(–0.110)	
H4a: Extrinsic maladaptive rewards → (–) Protection motivation	0.041		0.140		0.003	
H4b: Extrinsic maladaptive rewards → (–) Past PMBs	0.045		0.075		0.051	
H6a: Threat severity → Protection motivation	0.062		(–0.067)		0.227*	
H6b: Threat severity → Past PMBs	0.101*		0.054		0.153*	
H8: Threat severity → Fear	0.290***		0.257**		0.323***	
H9a: Fear → Protection motivation	0.086		0.122		(–0.023)	
H9b: Fear → Past PMBs	0.035		0.010		0.022	
H10a: Response efficacy → Protection motivation	0.236***		0.063		0.381***	
H10b: Response efficacy → Past PMBs	0.441***		0.305***		0.562***	
H12a: Response costs → (–) Protection motivation	(–0.190)*		(–0.183)		(–0.224)**	
H12b: Response costs → (–) Past PMBs	(–0.104)		(–0.110)		(–0.139)	
<i>Rival Explanations and Demographic Controls</i>						
Management support → Protection motivation	0.373***		0.371***		0.308**	
Management support → Past PMBs	0.135*		0.139		0.041	
Managerial position → Protection motivation	0.116*		0.142		0.063	
Managerial position → Past PMBs	0.153***		0.191**		0.126*	
<i>Extensions to the Baseline PMT Model</i>						
H1a: SETA frequency → Intrinsic maladaptive rewards	0.026		0.082		0.118*	
H1b: SETA frequency → Extrinsic maladaptive rewards	0.110*		0.195**		0.070	
						(continues)

Table 3. Continued

Tested paths	Overall		Low org. commit.		High org. commit.	
	B		B		B	
H1d: SETA frequency → Threat severity	0.180***		0.213**		0.171**	
H2a: SETA frequency → Response efficacy	0.316***		0.288***		0.270***	
H2c: SETA frequency → Response costs	0.078		0.096		0.032	
†Intrinsic maladaptive rewards → Response costs	0.301***		0.485***		0.175*	
†Extrinsic maladaptive rewards → Response costs	0.142*		(-0.078)		0.152	
†Response efficacy → Response costs	(-0.392)***		(-0.309)***		(-0.363)***	
<i>Variance explained</i>						
Intrinsic maladaptive rewards	0.001		0.007		0.014	
Extrinsic maladaptive rewards	0.012		0.038		0.005	
Threat severity	0.032		0.045		0.029	
Response efficacy	0.100**		0.083		0.073	
Response costs	0.249***		0.328***		0.176*	
Fear	0.084*		0.066		0.105*	
Protection motivation	0.351***		0.264**		0.556***	
Past PMBs	0.331***		0.238**		0.452***	

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; † = relationships suggested by modification indices.

model fit and equation-level model fit; (4) we not only included protection motivations, but also assessed previously performed protective behaviors via past PMBs. Our theoretical extensions include the further contributions of (5) incorporating organizational commitment as a key moderator of the PMT-based appraisals and (6) proposing SETA programs as potential antecedents to PMT.

As one of our theoretical extensions, we introduced SETA programs as an antecedent to the base PMT model. SETA programs are essential to securing organizational information. Our results clearly indicate that SETA programs significantly increase components of both threat and coping appraisal, meaning that SETA programs can act as the fear appeals process that is central to PMT. In particular, these programs increase threat severity awareness and response efficacy levels. Although SETA programs have previously been shown to be useful (e.g., [24]), we are the first to demonstrate a direct link between SETA, PMT, and the fear appeal process. By definition, SETA programs should explain threats and appropriate coping procedures within organizations. As noted above, fear appeals should perform exactly the same tasks of notifying employees of threats, threat severity, and appropriate coping behaviors.

We offer two additional new SETA-related results that should prove fruitful to both researchers and practitioners. We discovered that SETA programs significantly increase extrinsic maladaptive reward perceptions. This could be a negative result for the organization, because although the education process increases security awareness, it also improves insiders' awareness that illicit financial gains can be achieved by exploiting organizational weaknesses. SETA program developers must understand this risk and design their programs to discourage extrinsic maladaptive rewards. A second new finding is that current SETA programs do not significantly influence insiders' view of response costs. The difficulty is that response costs often conflict with individual or organizational goals, such as trying to get one's own assignment completed on time. This problem occurs frequently in organizational security processes. It is vital that insiders understand why they need to perform the PMBs in spite of any perceived personal costs associated with them. SETA programs should emphasize the need to endure these response costs for the organization's sake after considering the organizational trade-offs.

For the base PMT model, components of the threat and coping appraisals as suggested by PMT explained variance in insiders' protection motivation and past PMB activity significantly. Overall, intrinsic maladaptive rewards, response efficacy, and response costs were strongly related to the insiders' protection motivation levels, and intrinsic maladaptive rewards, threat severity, and response efficacy significantly explained previous PMBs. Of these, response efficacy was the most significant driver (per magnitude of standardized path weights) of insiders' motivation to protect their organizations from information security threats as well as their protective behaviors. This finding underscores the likelihood that the coping appraisal process is more vital to increasing protection motivation and protective actions than the threat appraisal process, as shown in other studies [67, 83]. For example, a PMB likely to carry a high response cost is that insiders should inform their superiors

quickly if they encounter incidents that run counter to the security guidelines. Unfortunately, the source of this detrimental activity is often an insider's coworker, who might engage in retaliation. For insiders to engage in the recommended response and inform their superiors of the issue, response costs could be minimized if the insiders' organizations guarantee the confidentiality of insiders' critical communications with authorized personnel. Further strategies for avoiding retaliation and increasing reporting could follow suggestions for practices related to whistleblowing (e.g., [57]). As expected, intrinsic maladaptive rewards were found to be significant in negatively predicting protection motivation and past PMBs.

Additionally, we provide evidence that in the context in which respondents are asked to rate their overall security threats and responses (as opposed to responding to one specific threat), insiders are generally not motivated by threat characteristics and fear, but that these effects differ with respect to organizational commitment levels. For example, threat severity and fear do not exhibit significant relationships with protection motivation when assessed in a general sense; however, threat severity becomes significant in the presence of high organizational commitment levels ($\beta = 0.227^*$). In the case of past PMBs, threat severity exhibits a significant relationship in the overall data set ($\beta = 0.101^*$) and the high organizational commitment subgroup ($\beta = 0.153^*$) but not the low-commitment subgroup ($\beta = 0.054$). Conversely, fear approaches but does not reach significance in low organizational commitment situations with respect to protection motivation levels. These findings are crucial because previous research has identified these factors—especially fear—as major driving forces in personal protective behaviors. Attempting to scare insiders about potential threats through messages that specifically attempt to elicit fear might be ineffective in organizations in which low organizational commitment is present among employees, even though it is potentially effective in other settings.

Our research also highlights the need to examine the interplay among PMT components. Specifically, our efforts demonstrate how perceptions of both forms of maladaptive rewards and response efficacy influence perceptions of response costs, thereby linking threat and coping appraisals—a relationship not noted in Rogers's [85] explication of PMT. Given that perceptions of response costs are the result of a cognitive cost-benefit analysis [83], maladaptive rewards and response efficacy are at the heart of that trade-off assessment, and response efficacy's strong influence on response cost perceptions is relatively constant across both low- and high-commitment environments. Furthermore, because maladaptive rewards and response costs are the least examined PMT components in the information security context [102], our results should be of value to information security researchers.

Our final addition to the PMT model is the introduction of organizational commitment as a key moderator. Our results should be of interest to researchers and practitioners alike. As shown in Table 3, the usefulness of the PMT model was vastly changed by the introduction of organizational commitment. In fact, these results should cause organizations to rethink their hiring, retention, and SETA processes. First, SETA programs significantly increased extrinsic maladaptive rewards for insiders with low organizational commitment ($\beta = 0.195^*$), whereas

this construct was insignificant for insiders with high organizational commitment ($\beta = 0.070$). The implication is that insiders with low organizational commitment are looking for rewards and opportunities without concern for their organization, and SETA programs do little to discourage them. In fact, these programs appear to expand some insiders' understanding of the benefits for not protecting the organization—insiders who exhibit limited commitment to their organizations.

Second, response costs are influenced largely by intrinsic maladaptive rewards for the low-commitment group ($\beta = 0.485^{***}$) whereas these costs were not as strongly influenced by intrinsic rewards in the high-commitment group ($\beta = 0.175^*$). We believe that this finding further underscores the fact that those with low organizational commitment levels are focused on personal benefits rather than organizational protection. When this relationship is combined with the other findings related to commitment's moderating effect, these discoveries are important for hiring and retention decisions and the development of formal SETA programs. The real impact becomes evident when one considers that only in the PMT model with high organizational commitment was there a significant relationship between threat severity, response efficacy, and response costs with protection motivation. Ultimately, this finding means that only highly committed employees were responding adaptively to the threat and coping appraisal processes. This is a major finding with regard to information security in today's environment, characterized by the dynamic nature of IT and the constant evolution of security risks. The challenge of building organizational commitment among employees with the goal of securing an organization is daunting, because it requires not only a good SETA program but also an effort to create a highly committed workforce. Our results indicate that when insiders lack commitment, they will care little about and act less upon the information security threats their organizations face on a daily basis. They are less adaptive and more apathetic than highly committed employees. This finding lends support to the theory that insiders committed to their organizations view organizational security threats as threats to themselves and are thus motivated to engage in protective actions to counter those threats.

In short, this study contributes to both research and practice, including the validation of a global security behavior construct (PMBs), the application of SETA as an antecedent to the PMT model, the application of organizational commitment as a moderator to the PMT model in organizational information security contexts, and the relationship between maladaptive rewards and response costs within PMT.

Limitations and Future Research

This study has several limitations that indicate compelling research opportunities. First, we obtained our data from a cross-sectional research panel in the form of a survey. Although this approach enhances realism and generalizability across different kinds of organizations and professionals, it does not allow us to establish causation and makes it difficult if not impossible to determine protection

motivation's true influence on adaptive behaviors following the initial formation of threat and coping appraisal components. Future research could thus extend our work by testing smaller portions of the model in controlled experiments in an effort to assess the validity of our model's causal mechanisms.⁴ In addition, our use of SETA frequency as a categorical variable limits the scope and details that can be reliably associated with SETA programs. Future research can provide greater insight into the impact of SETA in the PMT theoretical model by considering richer variations of SETA manipulations such as training length, use of fear appeals, use of media, follow-up sessions, and so on. Although our study is a useful starting point, it is possible that the components of PMT are influenced differently relative to the type of SETA manipulation. The quality and quantity of SETA programming are also prominent issues, and we believe that both should be examined for their separate and perhaps interactive effects.

Moreover, our results regarding fear should not be used to rule out categorically the potential role of fear in PMT. First, Rogers [86] has brought fear back into PMT, and meta-analysis has confirmed that fear plays a role in PMT apart from the threat itself; thus, researchers should not be too quick to dismiss it [9, 31]. We posit that the salience of fear depends on how specific the context is for an individual behavior (e.g., virus protection) versus a broad set of behaviors (e.g., PMBs and SETA programs). Fear likely lacked salience in our context because we relied only on the SETA programs to which the respondents were exposed; if the threats from failure to protect their organizations were not sufficiently strong or salient, then fear would not be very strong. Hence, future security-based PMT research needs to examine more closely both fear appeals and threats themselves and their relationships with fear. This has been recently successfully performed in [9] for specific fear-appeal manipulations (e.g., conducting backups), but not for SETA programs or organizational commitment. Such research needs to be further tested in terms of fear appeal manipulations through controlled SETA programs and accounting for organizational commitment.

Second, we assessed past PMBs via respondents' self-reports of their protective activities. Because of organizations' general unwillingness to provide academicians access to employees for research purposes, especially when the research focuses on organizational information security matters [47], self-reports afford academicians one method of obtaining data to model and understand important insider activities as they relate to organizational information security. Despite this opportunity, self-reports are among a variety of other data collection techniques, which offer complementary advantages to one another. For example, third-party ratings (e.g., supervisor, co-worker), business process data, and interviews, when made available, may be used to further assess insiders' behaviors [104]. We thus call on future behavioral information security research that leverages these additional techniques.

There is also an interesting association between job satisfaction and organizational commitment that should be further explored in future research. Our study showed a high positive correlation (0.763) between the two. It makes sense that these would be highly related, because it is difficult to imagine high commitment to an organization

paired with low job satisfaction. Nonetheless, these are distinct constructs not only in our study but also in the literature. Even with a high correlation, the two constructs fail to share more than 40 percent of variance between them. Hence, there is an exciting opportunity to further examine and build on these relationships in order to develop greater insight into the role of job satisfaction in organizational security compliance.

Finally, our tests are based on data from U.S.-based professionals. Although management support displayed a significant influence on insiders' protection motivation, cultural differences could moderate these important relationships. Such differences have been found in technology settings between collectivistic and individualistic cultures, and between high and low power distance cultures [55, 78]. Compared to individualists, collectivists tend to be more cooperative and oriented toward the good of the many. Those in high power distance cultures tend to be more willing to obey and not question authority than those in low power distance cultures. Thus, insiders from collectivist and high power distance cultures, such as many Arab countries, China, and India, might be influenced differently by the antecedents and rival hypotheses of our model. These possibilities have yet to be examined fully in an IS PMT security context.

Conclusions

To further illuminate why insiders act to protect their organizations from information security threats, we have shown the importance of often-forgotten factors associated with PMT—namely, maladaptive rewards and response costs—in the appraisal processes as they relate to protection-motivated behaviors (PMBs). We have also detailed how insiders' organizational commitment levels moderate much of the processes indicated by PMT; hence, organizational commitment is instrumental in making organizational information security threats personally relevant to insiders. Finally, organizational SETA efforts were shown to bolster components in both threat and coping appraisals.

Supplemental File

Supplemental data for this article can be accessed on the publisher's website at <http://dx.doi.org/10.1080/07421222.2015.1138374>

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NOTES

1. We note that other research exists on similar topics, such as identification and internalization; however, the three-component view of organizational commitment developed in [62] is the most widely evaluated model of employees' connections and personal affiliations with organizations [2, 66].

2. Prior to the collection of data regarding perceptions of information security threats, potential responses, and other PMT-based constructs, respondents were asked to read a short statement about types of security threats and to reflect on their and others' previous experiences with them in workplaces: "You will be asked a series of questions regarding information security threats in your organization. Organizations face many threats to information, including but not limited to spyware and malware, external hackers attempting to gain access to databases housing important data, and even coworkers who for one reason or another choose to use their access for malicious purposes. Accordingly, no organization is completely immune to these threats. We ask you to think about these threats as well as any previous experiences you or those around you have had with these dangers as you respond to the following set of questions."

3. Because we used the MLR estimator in Mplus, the statistics listed (e.g., the scaling correction factors) allow us to assess differences in χ^2 scores via the Bryant and Satorra [10] approach.

4. Some research indicates that previously performed behaviors may be used as a substitute for behaviors, which would occur in the future; however, much research is needed to determine the validity of these arguments, and for this purpose in an organizational information security context, we provide Appendix 3 [online supplemental data].

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